



Study on the influence of railing ventilation rate on the aerodynamic mechanism of vortex-induced vibration in double-layer steel truss bridge

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ABSTRACT

In analyzing aerodynamic oscillations of extended-span bridges, vortex-induced vibration (VIV) emerges as a pivotal element in designing for wind resistance. As an essential auxiliary component of the main girder, bridge deck railings alter the girder's aerodynamic properties when their ventilation rate is reduced, significantly affecting the VIV response of long-span bridges. To explore how railing ventilation rate on the VIV characteristics of double-layer steel truss girder sections and their underlying mechanisms, this study varied the ventilation rates of maintenance walkway railings, upper crash railings, pedestrian railings, and lower crash railings, establishing 13 test conditions for wind tunnel experiments and numerical simulations. Findings indicate that decreasing ventilation rate in maintenance walkway or upper crash railings intensifies the stagnation of inflow at the upper deck's forward edge, inducing flow separation at the railing tops and stronger vortex shedding, thereby amplifying the vertical and torsional VIV responses of the main girder. Notably, when the upper crash railing ventilation rate is 0 %, vertical VIV excitation is most significant, characterized by a 33 % maximum amplitude increase. Reducing the ventilation rate of pedestrian or lower crash railings suppresses the formation and shedding of vortex structures in the wake region of the lower bridge deck, attenuating wake vorticity and aerodynamic asymmetry, thereby effectively reducing torsional VIV responses. A pedestrian railing ventilation rate of 0 % yields optimal suppression, achieving a secondary torsional VIV suppression rate of 78.5 % and a primary torsional VIV suppression rate of 71.3 %.

1. Introduction

With the continuous advancement of global transportation infrastructure construction, steel truss bridges are increasingly widely used in complex geomorphological environments such as deep canyons due to their structural characteristics. However, the wind fields prevalent in such bridge locations often exhibit complex spatial distribution characteristics, notably high-velocity winds and intense turbulence. The aerodynamic impacts of these conditions frequently surpass the design parameters stipulated in current wind-resistant codes, thereby presenting formidable technical challenges to both bridge design and construction [1–3]. Double-deck steel truss suspension bridges are extensively employed due to their superior traffic capacity and efficient use of alignment resources. However, their complex cross-sectional morphology, along with a multitude of discrete components such as

deck railings [4], gives rise to diverse aerodynamic characteristics under turbulent flow conditions. These characteristics markedly affect airflow separation and reattachment, which in turn complicates the vortex-induced vibration (VIV) performance of such double-deck steel truss structures [5]. This not only presents potential hazards to pedestrian comfort and vehicular operational safety but can also compromise the long-term serviceability and durability of the structure [6]. For road-rail dual-purpose double-deck steel truss bridges, ensuring smooth train operation and passenger comfort is of paramount importance; consequently, these structures are subject to more stringent requirements for VIV control. In this context, the influence of railing ventilation rate—a critical factor affecting the main girder's aerodynamic performance—on the VIV characteristics of double-deck steel truss girders, along with the underlying mechanisms, warrants thorough investigation.

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Existing studies have revealed the significant impact of varying railing ventilation rates on the VIV performance of the main girder. For instance, research by Bai et al. [7] indicated that closed railings incorporating vertical intervals are effective in suppressing its vertical bending VIV, whereas continuously arrayed closed railings, devoid of such vertical gaps, demonstrate superior efficacy in mitigating its torsional VIV. Through wind tunnel experiments, Cui et al. [8] showed that reducing the ventilation rate of lateral railings effectively suppresses VIV, with torsional and vertical VIV exhibiting different sensitivities. Specifically, a ventilation rate of 25 % effectively suppresses vertical VIV, while 33 % effectively suppresses torsional VIV. Liu et al. [9] explored how enclosing various railings affects the main girder's vertical VIV, finding that enclosing pedestrian and median railings moderately reduces VIV amplitude. Wu et al. [10] found that ventilated railings, compared to enclosed railings, significantly suppress both vertical and torsional VIV, markedly reducing VIV amplitude. Liu et al. [11] further revealed that sealing the upstream railings reduces variable wind pressure on the girder surface, interrupts vortex development on upper and lower surfaces, and reduces VIV occurrences. Yao et al. [12] found that the presence of railings significantly amplifies the VIV response of steel truss beam bridges, but reducing the ventilation rate of railings effectively mitigates VIV amplitude. Yan and Yagi et al. [13] performed wind tunnel experiments to evaluate VIV impacts on box girders with non-perforated and perforated railings. Their findings indicate that VIV amplitude decreases as the porosity of the railings increases. For non-perforated railings, the amplitude reduces with an increase in the number of horizontal bars and the aspect ratio. Conversely, in perforated railings, an increase in the number of vertical bars is detrimental to VIV suppression. Li et al. [14] determined that the ventilation rate of traffic railings significantly affects torsional VIV. Spaced sealed railings effectively suppress torsional VIV responses but increase aerodynamic drag, adversely impacting driving comfort and bridge aesthetics.

While the aforementioned studies provide a foundational understanding of bridge VIV characteristics and the influence of railing ventilation rates, notable limitations persist. Primarily, current research has largely focused on single-deck main girder cross-sections, with systematic investigations into the VIV characteristics of double-deck steel truss girder cross-sections remaining relatively scarce. Furthermore, the existing research is mostly limited to the analysis of the vibration suppression effect of the railing ventilation rate, and the in-depth analysis of the aerodynamic mechanism of the relevant airflow separation, vortex structure evolution and wake characteristics is not sufficient. Particularly in double-deck steel truss girders, the influence of variations in the lower deck railing ventilation rate on the overall flow field is subject to modulation by complex aerodynamic interference from the upper bridge deck. This interaction can potentially enhance flow field asymmetry, thereby rendering the mechanisms of VIV even more intricate. To date, such specific issues have not been subjected to in-depth investigation in the relevant literature.

Within the complex wind environment of the bridge site investigated in this study, both $+3^\circ$ and $+5^\circ$ demonstrate a high probability of occurrence. Notably, the $+3^\circ$ is not only prevalent for many bridge structures but also possesses considerable representativeness and practical engineering applicability in the field of bridge wind-resistance research. Given that the primary objective of this study is to systematically investigate the influence of different railing types and their ventilation rates on the VIV characteristics of double-deck steel truss girders, and to conduct an in-depth analysis of the associated aerodynamic mechanisms—rather than to undertake an exhaustive examination of responses across all possible angles of attack. Therefore, within the constraints of limited computational resources and the project timeline, selecting the $+3^\circ$ for in-depth analysis offers prominent academic value and practical significance, particularly by effectively elucidating VIV mechanisms, enabling comparative validation against existing findings, and providing crucial references and directional

guidance for future analogous research.

Therefore, this study takes a double-deck steel truss girder suspension bridge with a main span of 765 m as the engineering background. Through the combination of wind tunnel test and numerical simulation, under the condition of $+3^\circ$, the influence of different types of railings on the vertical bending and torsional vortex vibration response of the double-deck steel truss girder section under various railing ventilation rate conditions is systematically studied, and the VIV characteristics and aerodynamic formation mechanism are discussed in depth, so as to provide theoretical basis and technical support for the wind resistance design of the bridge.

2. Engineering background

This study focuses on a double-tower, three-span continuous steel truss suspension bridge with a main span configuration of 230 m + 765 m + 230 m. The main girder adopts a double-layer steel truss design, with a total width of 36.5 m and a height of 15.75 m. Located within a mid-subtropical humid monsoon climate zone, the bridge is significantly influenced by monsoonal winds. With its long span and exposure to harsh river wind conditions, the wind field exhibits complex characteristics, making wind-resistant design particularly critical. The standard cross-sectional profile of the primary girder appears in Fig. 1.

3. Segment model wind tunnel testing

3.1. Segment model design

Testing occurred in Chongqing University's direct-flow wind tunnel laboratory, with wind speeds spanning 0.5–35.0 m/s and consistent inflow conditions. To accurately replicate the aerodynamic profile of the actual bridge, a bridge segment model with a scale ratio of 1:55 was designed and fabricated, considering constraints such as wind tunnel conditions, model dimensions, and mass [15]. The model's girder measures 0.286 m in height and 0.569 m in width, as shown in Fig. 2.

Fig. 3 illustrates the spring suspension system of the bridge segment model. The segment model is suspended on an angle-of-attack turntable. Suspension is achieved through eight springs attached to steel brackets, allowing vertical translation and torsional rotation about its sectional axis. This setup ensures accurate simulation of the vibrational characteristics in vertical bending and torsion with two degrees of freedom [16].

In the experiment, to prevent the spring suspension system from interfering with the flow field, it was positioned outside the wind tunnel [17]. Two laser displacement sensors were employed to measure the displacement response of the segment model. Counterweight mass and placement were modified to attain the necessary system mass and inertial moment, while spring rigidity and positioning were calibrated to achieve suitable. The primary design parameters of the segment model are listed in Table 1.

3.2. VIV response of the original section

To evaluate the VIV performance of the original section, aeroelastic tests on the model were conducted in a wind tunnel with uniform flow. The tests were performed at three wind attack angles: 0° , $+3^\circ$, and -3° . The results are presented in Fig. 4.

The test results presented in Fig. 4 indicate that significant VIV of the model was observed only at $+3^\circ$. The lock-in range for the vertical bending VIV corresponded to a full-scale wind speed of 4.93–6.72 m/s, with a maximum amplitude of 75.54 mm. The torsional VIV included two lock-in regions, corresponding to full-scale wind speed ranges of 8.13–9.52 m/s and 12.73–18.06 m/s, with maximum amplitudes of 0.079° and 0.181° , respectively. In contrast, no obvious VIV phenomena were observed at -3° and 0° .

To further analyze the vibration characteristics of the model during

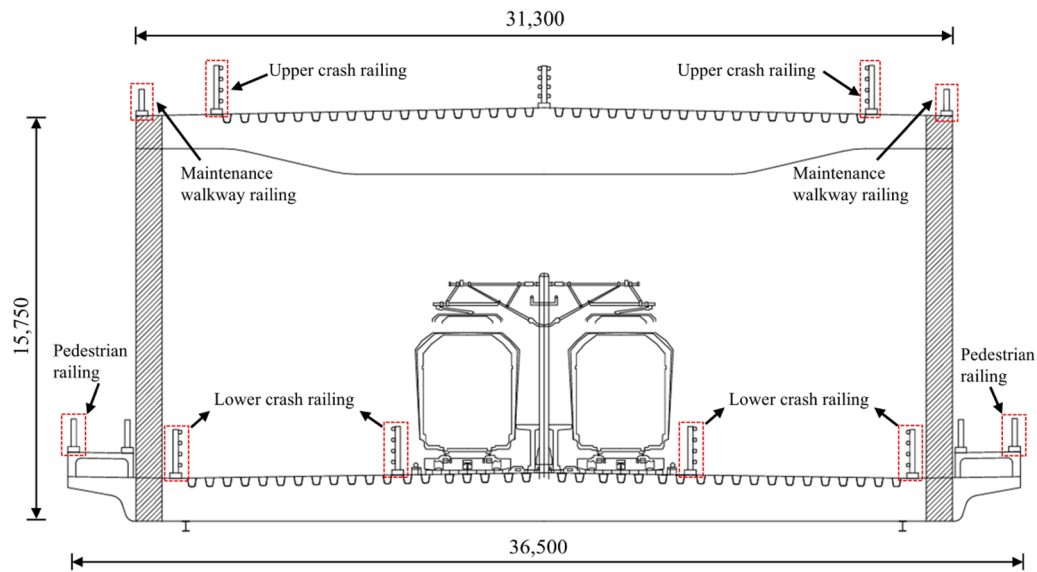


Fig. 1. Standard cross-section of the primary girder (Unit: mm).



(a) overall structure



(b) local details

Fig. 2. Bridge segment model.

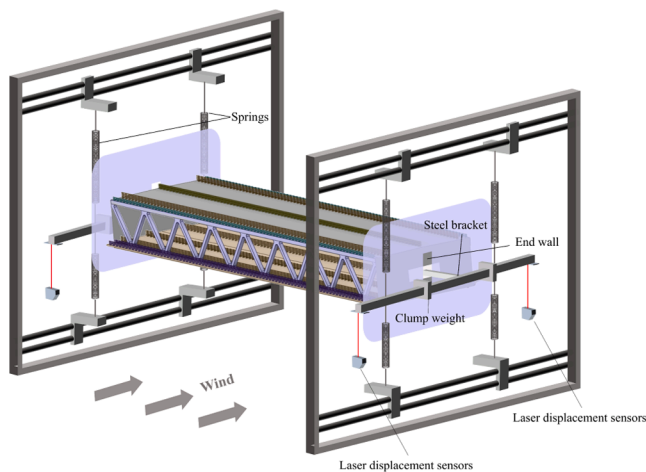


Fig. 3. Spring suspension system of the segment model.

Table 1

Main design parameters of segment model.

Parameter	Actual bridge design value	Similarity ratio	Segment model design value
Equivalent mass (m)	$6.845 \times 10^4 \text{ kg/m}$	1:55 ²	22.628 kg/m
Equivalent moment of inertia (I_m)	$1.309 \times 10^7 \text{ kg}\cdot\text{m}^2/\text{m}$	1:55 ⁴	1.431 kg·m ² /m
Vertical bending fundamental frequency (f_h)	0.211 Hz	1:10.51	2.218 Hz
Vertical bending damping ratio (ζ_h)	0.50 %	1:1	0.38 %
Torsional fundamental frequency (f_a)	0.478 Hz	1:10.53	5.033 Hz
Torsional damping ratio (ζ_a)	0.50 %	1:1	0.41 %

the tests, a spectrum analysis of the VIV displacement time-history data was performed using the Fast Fourier Transform (FFT). Fig. 5 illustrates the displacement time-series plots and corresponding amplitude spectra

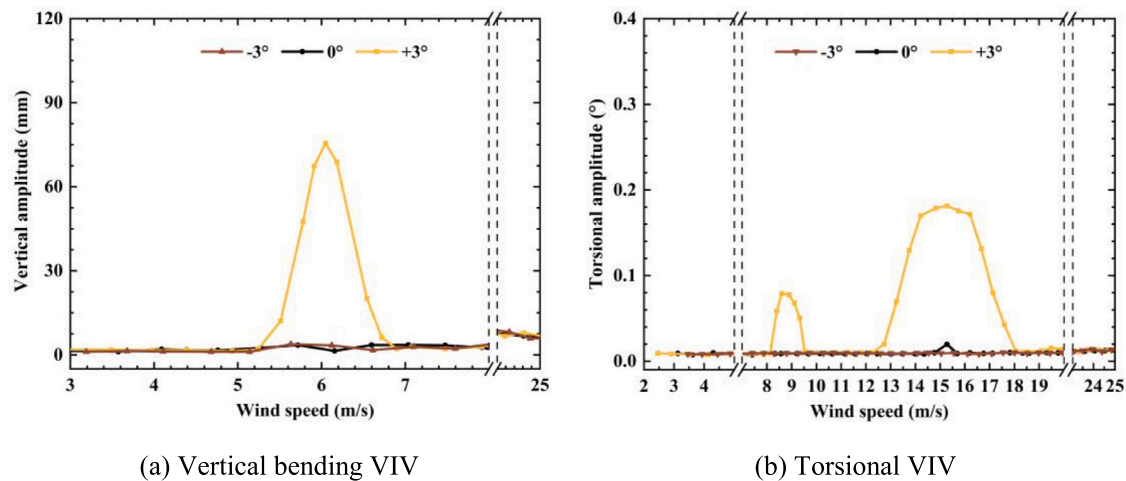


Fig. 4. Results of VIV of original section in wind tunnel tests.

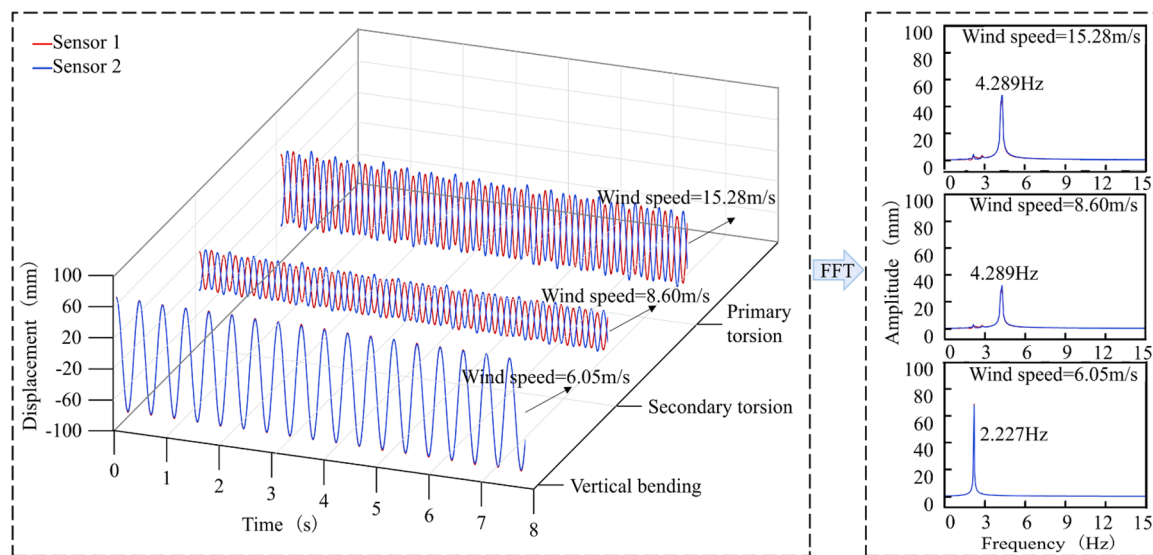


Fig. 5. Time-series and Spectral Analysis of the Peak VIV Response at $+3^\circ$.

recorded by two laser displacement sensors under $+3^\circ$, obtained at the main girder section's peak VIV response locations in the completed bridge state.

As shown in Fig. 5, at $+3^\circ$, the peak response of the vertical VIV occurred at a full-scale wind speed of 6.05 m/s. The displacement time-series recorded by the two laser displacement sensors were in-phase, with a dominant frequency of 2.227 Hz, which is nearly identical to the model's vertical natural frequency. In contrast, the peak responses of the torsional VIV occurred at a full-scale wind speeds of 8.60 m/s and 15.28 m/s, forming two separate lock-in regions. Within both lock-in regions, the displacement time-series were anti-phase, and the dominant frequency was 4.289 Hz in both cases, which is in high agreement with the model's torsional natural frequency.

To further reveal the aerodynamic generation mechanism, the St [18] corresponding to each response peak were calculated. The results show that at $+3^\circ$, the St number for the vertical bending VIV peak was 0.47, while the St numbers for the two torsional VIV peaks were 0.64 and 0.36, respectively. This difference clearly indicates that the vertical bending and torsional VIV of the main girder section are driven by vortices with different shedding frequencies. Specifically, the excitation mechanism for the vertical bending VIV is relatively singular, caused by a single, dominant, and stable vortex shedding pattern. By contrast, the

excitation mechanism for torsional VIV is more complex. It is speculated that due to the section's unique geometry, the torsional VIV successively locks into two different vortex shedding modes at different wind speed ranges. These modes correspond to different St numbers, thus forming the observed "secondary" and "primary" independent lock-in regions.

3.3. Experimental condition

To investigate the influence of bridge deck railing ventilation rates on the VIV response of the main girder section under $+3^\circ$, the ventilation rates of bridge deck railings (including maintenance walkway railings, upper crash railings, lower pedestrian railings, and lower crash railings) were adopted as the primary research parameter. Thirteen combinations of railing ventilation rates were tested in wind tunnel experiments, with specific experimental conditions detailed in Table 2. Based on the original ventilation rate of the main girder section in the completed bridge state, the railing ventilation rate was progressively reduced to 0 % by adding horizontal bars. The ventilation rate configurations for the four types of railings and the layout of bridge deck facilities are illustrated in Fig. 6.

Table 2
Wind tunnel test conditions.

Railing type	Condition number	Ventilation rate	Test model
Maintenance walkway railing	JXD-YS	62.5%	
	JXD-1	43.8%	
	JXD-2	0.0%	
Upper crash railing	SFZ-YS	57.1%	
	SFZ-1	35.7%	
	SFZ-2	0.0%	
Pedestrian railing	RXD-YS	66.7%	
	RXD-1	50.0%	
	RXD-2	33.3%	
	RXD-3	0.0%	
Lower crash railing	XFZ-YS	57.1%	
	XFZ-1	35.7%	
	XFZ-2	0.0%	

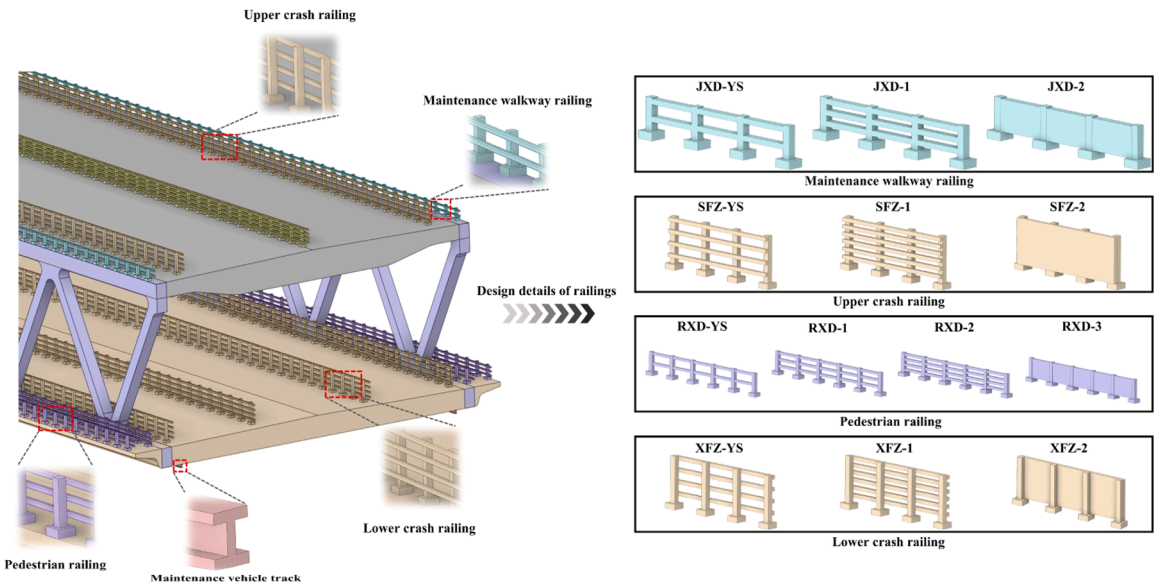


Fig. 6. Layout diagram of bridge deck facilities.

3.4. Wind tunnel experiment results

3.4.1. Test results for maintenance walkway railings with varying ventilation rates

Under $+3^\circ$, the vertical bending and torsional VIV responses for the YS, JXD-1, and JXD-2 conditions exhibit similar trends, as illustrated in Fig. 7.

The maximum vertical bending VIV amplitudes for the three conditions are 75.54 mm, 78.08 mm, and 90.08 mm, respectively. The maximum amplitudes for secondary and primary torsional VIV were found to be 0.079° , 0.081° , and 0.092° , and 0.181° , 0.188° , and 0.212° , respectively. Compared to the YS condition, the vertical bending VIV amplitudes for the JXD-1 and JXD-2 conditions increase by 3.4 % and 19.3 %, respectively. The maximum secondary torsional VIV amplitudes rise by 2.1 % and 16.9 %, respectively, while the maximum primary torsional VIV amplitudes increase by 4.0 % and 17.1 %, respectively. With maintenance walkway railings at 0 % ventilation rate, mitigation of vertical and torsional VIV is least effective, suggesting that lower ventilation rate in these railings intensifies girder section VIV responses.

3.4.2. Test results for upper crash railings with varying ventilation rates

Under $+3^\circ$, the vertical bending and torsional VIV responses for the YS, SFZ-1, and SFZ-2 conditions are illustrated in Fig. 8. Compared to the YS condition, the vertical bending and torsional VIV responses of the main girder under the SFZ-1 and SFZ-2 conditions consistently increase, reaching their peak under the SFZ-2 condition. The maximum vertical bending VIV amplitude rose from 75.54 mm to 100.44 mm, secondary torsional VIV from 0.079° to 0.093° , and primary torsional VIV from 0.181° to 0.212° , with increases of 33.0 %, 18.0 %, and 17.2 %, respectively. These results indicate that reducing the upper crash railings ventilation rate significantly impairs the main girder's aerodynamic performance, markedly amplifying VIV amplitudes, particularly for vertical bending VIV responses.

3.4.3. Test results for lower pedestrian railings with varying ventilation rates

At $+3^\circ$, as the ventilation rate of pedestrian railings progressively decreases, the vertical bending and torsional VIV responses of the main girder continuously diminish, as shown in Fig. 9. When the pedestrian railing ventilation rate reaches 0 % (RXD-3 condition), the suppression of vertical bending and torsional VIV is optimal. The maximum vertical bending VIV amplitude decreased from 75.54 mm to 58.42 mm, the secondary torsional VIV maximum amplitude from 0.079° to 0.017° , and the primary torsional VIV maximum amplitude from 0.181° to 0.052° . The main girder's vertical bending VIV suppression effect was only 22.7 %, whereas the secondary and primary torsional VIV

suppression effects reached 78.5 % and 71.3 %, respectively. These results suggest that reduced ventilation in pedestrian railings suppresses torsional VIV far more effectively than vertical bending VIV.

3.4.4. Test results for lower crash railings with varying ventilation rates

As shown in Fig. 10, at $+3^\circ$, reducing the ventilation rate of lower crash railings continuously decreases the vertical bending and torsional VIV responses of the main girder, contrasting with the excitation effects of maintenance walkway and upper crash railings.

For XFZ-1 and XFZ-2 conditions, the maximum vertical bending VIV amplitudes are 65.73 mm and 61.44 mm, secondary torsional VIV amplitudes are 0.051° and 0.041° , and primary torsional VIV amplitudes are 0.107° and 0.065° , respectively. Compared to the YS condition, the vertical bending VIV amplitude under XFZ-1 and XFZ-2 conditions decreased by 13.0 % and 18.7 %, respectively, the secondary torsional VIV amplitude by 35.8 % and 48.2 %, and the primary torsional VIV amplitude by 41.0 % and 64.0 %. These results indicate that at 0 % ventilation rate of lower crash railings, the suppression of vertical bending and torsional VIV is optimal, with torsional VIV suppression significantly outperforming vertical bending VIV suppression.

4. Numerical simulation of the main girder section

4.1. Solution setup

Two-dimensional CFD simulations, characterized by efficiency and simplicity, are widely used to evaluate bridge aerodynamic performance [19]. Based on simplified principles—identical aerodynamic shapes of bridge components, equal total wind-affected area in the vertical direction, and similar aerodynamic interference in the horizontal direction [20]—this study employs a two-dimensional CFD model. The computational geometry model retains a 1:55 scale, consistent with the wind tunnel model, featuring a domain of $32B \times 16B$ (B as model width) and a 0.5 % blockage ratio. The domain is divided into an inner region (near-wall rigid motion zone and dynamic mesh zone) and an outer region (background mesh zone [21]). The outer region employs structured meshes, while the inner region uses unstructured meshes [22]. The inlet is assigned a velocity inlet boundary condition [23], the outlet a pressure outlet boundary condition [24,25], the top and bottom boundaries are symmetry boundaries, and the double-layer steel truss girder section is a no-slip wall [26,27]. Fig. 11 depicts the division of the computational domain and its boundary condition setup.

Numerical simulations were conducted using Fluent software, with a mathematical model based on the Reynolds-averaged Navier-Stokes equations [28–30]. The SST $k-\omega$ turbulence model was employed for transient computations, with a turbulence intensity of 0.5 %. The

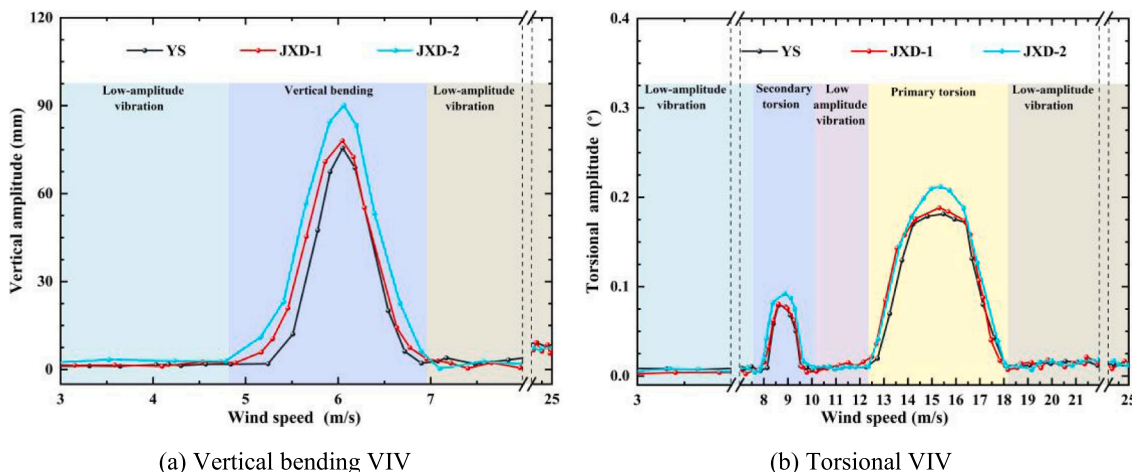


Fig. 7. Main girder VIV response of maintenance walkway railing at different ventilation rates.

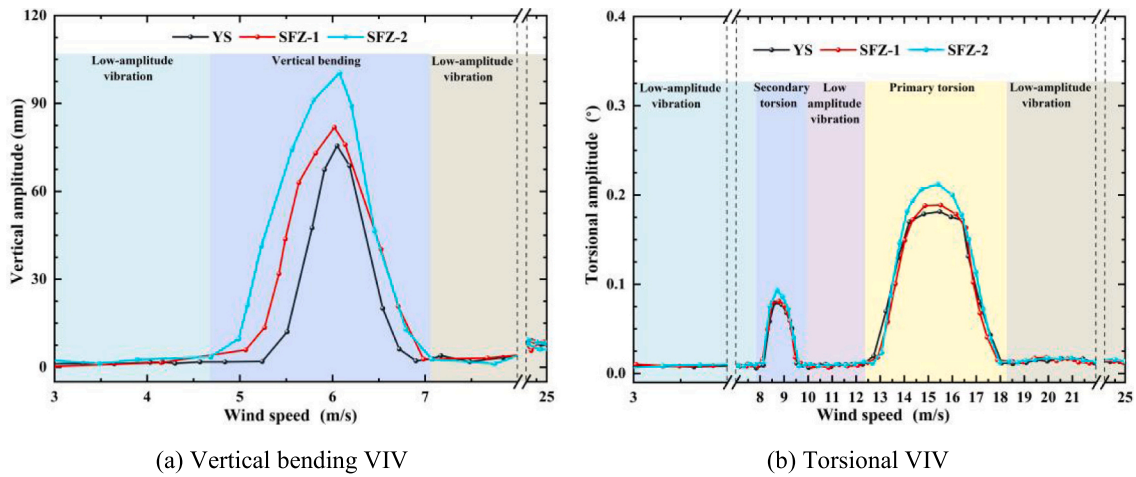


Fig. 8. Main girder VIV response of upper crash railing at different ventilation rates.

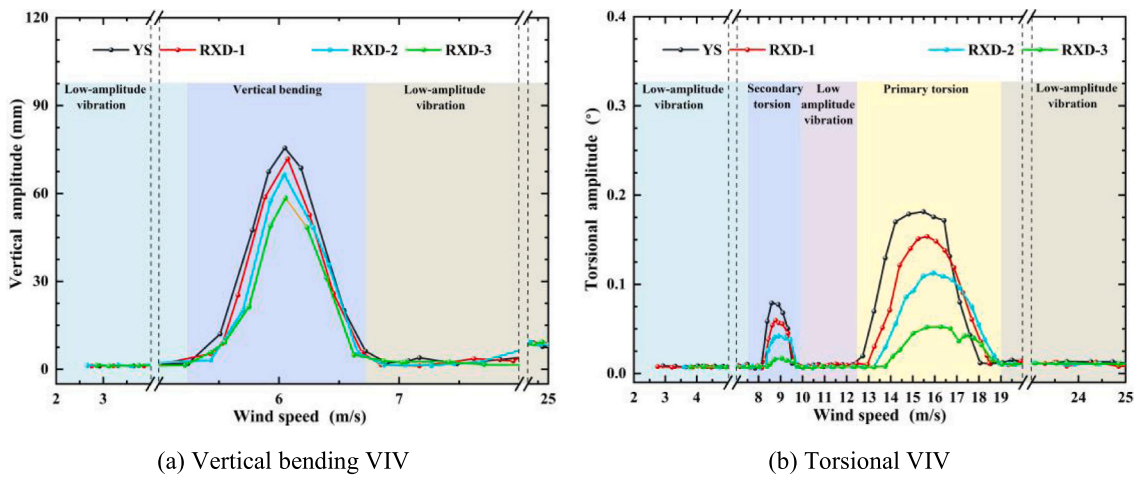


Fig. 9. Main girder VIV response of pedestrian railing at different ventilation rates.

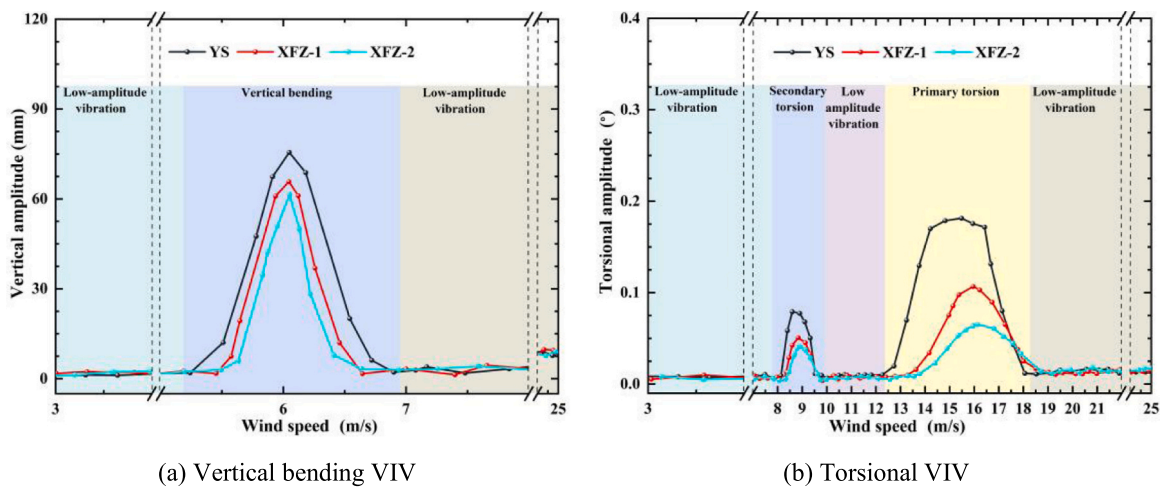


Fig. 10. Main girder VIV response of lower crash railing at different ventilation rates.

SIMPLE algorithm was selected for pressure-velocity coupling, and the residual convergence criterion was set to 1×10^{-5} .

4.2. Irrelevant verification

To ensure the validity and reliability of CFD results, time step and grid independence studies were conducted for the two-dimensional main girder section model at $+3^\circ$ and 10.0 m/s wind speed.

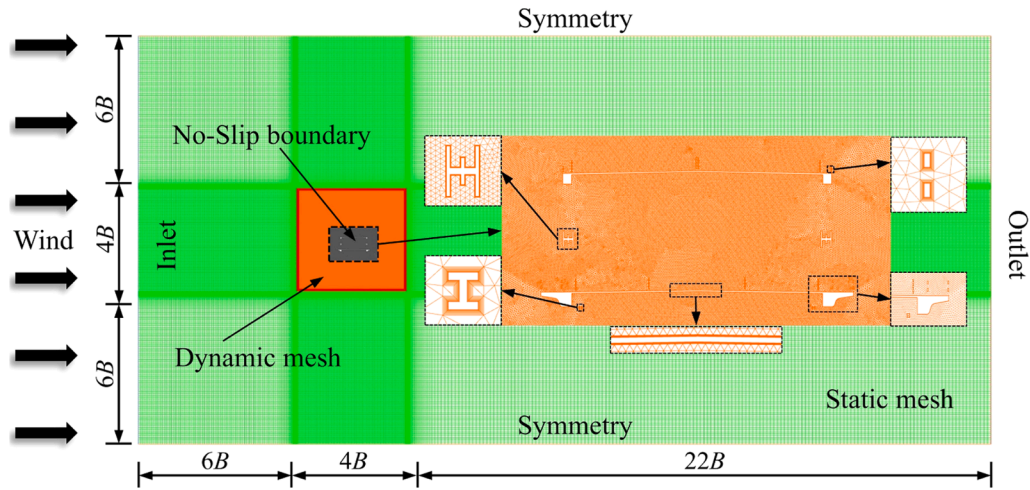


Fig. 11. Computational domain division and boundary condition setup.

- (1) Five time steps— 1×10^{-3} s, 5×10^{-4} s, 1×10^{-4} s, 5×10^{-5} s, and 1×10^{-5} s—were used to compute drag, lift, and moment coefficients. Time step independence was verified by comparing the mean and standard deviation of these aerodynamic force coefficients, with detailed results presented in Fig. 12.

Fig. 12 shows that mean and standard deviation values for time steps of 1×10^{-3} s and 5×10^{-4} s exceed those of three other groups significantly. Decreasing the time step below 1×10^{-4} s yields no notable result changes, so a 1×10^{-4} s step was chosen, with 1×10^5 steps over a 10 s computation period.

- (2) Four grid sizes— 39.3×10^4 , 78.5×10^4 , 98.0×10^4 , and 117.6×10^4 cells—were selected to compute the model's aerodynamic force coefficients. Grid independence was verified by comparing the mean and standard deviation of these coefficients across the four grid sizes, with detailed results shown in Fig. 13.

Fig. 13 shows that increasing the grid size beyond 78.5×10^4 cells yields no significant changes in results. Although finer grids enhance computational accuracy, they substantially increase computational cost. Thus, the numerical simulation grid size was set to 78.5×10^4 cells.

During the meshing process, to accurately capture the aerodynamic force distribution on the bridge surface, we partitioned the double-deck steel truss girder section into three regions: upper and lower decks, web

members, and railings, and refined the near-wall mesh in each region. The key metric for mesh quality is the dimensionless distance, y^+ . According to the SST $k-\omega$ turbulence model adopted in this study, the y^+ value of the first near-wall grid node must be less than 1 to ensure computational accuracy, as this allows for the accurate resolution of the viscous sublayer of the boundary layer. As shown in Fig. 14, the final y^+ [31] distribution verifies the reasonability of the mesh division. Although the y^+ values near the railings and web members are significantly higher than those near the decks, all y^+ values are less than 1, indicating that the mesh refinement meets the computational requirements and can ensure the reliability of the simulation results.

4.3. Reliability verification of numerical calculations

To confirm the precision and dependability of this study's computational modeling approach, the static aerodynamic force coefficients of the model were computed at a full-scale wind speed of 20.0 m/s for five wind angles of attack (-5° , -3° , 0° , $+3^\circ$, $+5^\circ$) and compared with force measurement test results, as shown in Fig. 15.

Fig. 15 shows that the aerodynamic force coefficients computed from numerical simulations at various wind angles of attack exhibit high agreement with experimental results, with a maximum error of 9.032 % and a minimum error of 1.234 %. The mean errors are 3.133 % for lift coefficient, 2.918 % for drag coefficient, 6.135 % for moment coefficient, and 4.062 % overall.

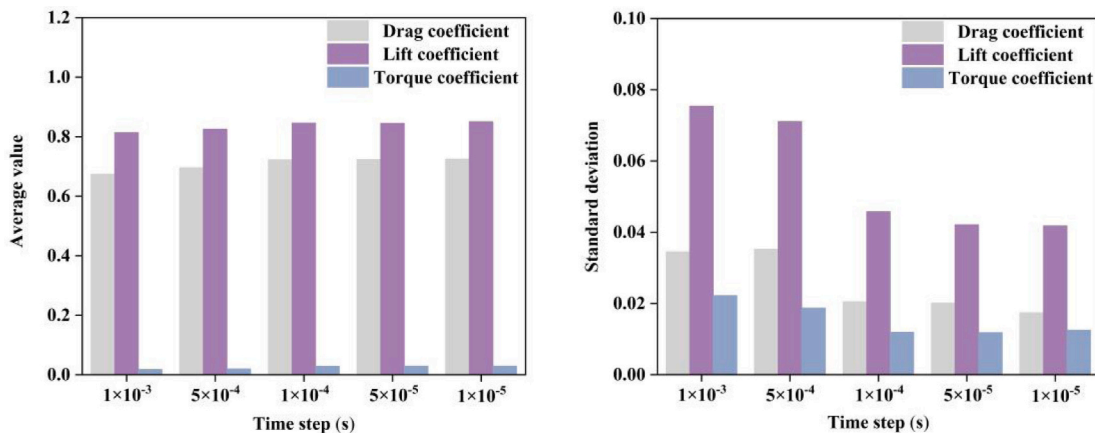
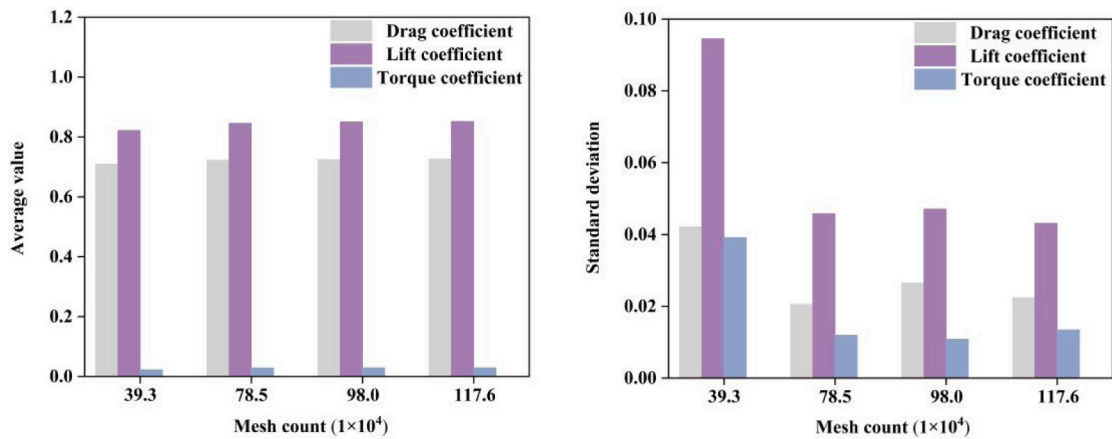


Fig. 12. Time step independence test.



(a) Three-component force coefficient average value (b) Three-component force coefficient standard deviation

Fig. 13. Mesh count independence test.

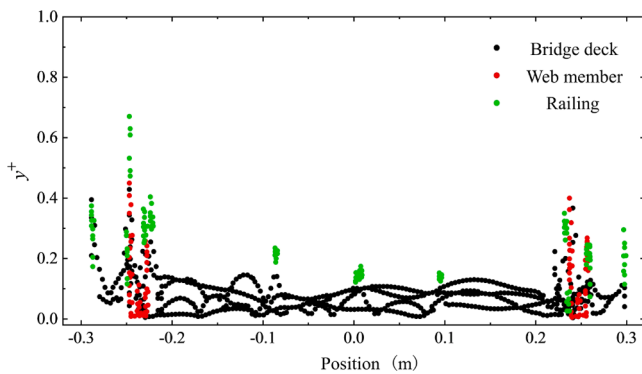


Fig. 14. Near-wall y^+ value distribution.

To further evaluate the accuracy of numerical simulations, the vertical bending VIV amplitudes at wind speeds corresponding to the vertical bending VIV lock-in range were computed for the model at $+3^\circ$ and $+5^\circ$ and compared with vibration test results, as shown in Fig. 16.

Fig. 16 shows that the vertical bending VIV amplitude errors between simulated and experimental values range from a maximum of 13.271 % to a minimum of -13.017 %. The mean amplitude errors are -4.052 % at $+3^\circ$ and -3.229 % at $+5^\circ$, with an overall mean error of -3.641 %. Compared to the -5.4 % error reported by Yao et al. [32] in a similar study, the accuracy is satisfactory, indicating that the two-dimensional model used in this paper has a high degree of computational reliability.

In summary, the numerical simulation results are generally larger than the wind tunnel test, and the reasons are speculated as follows: (1) The two-dimensional model used in the numerical simulation cannot fully capture the real three-dimensional characteristics of the main beam. In contrast, wind tunnel tests can restore the details of bridge deck components (such as railings or maintenance vehicle tracks) in more detail. These components may lead to an increase in the local turbulence intensity of the main girder section, which will have a certain inhibitory effect on the vortex-induced vibration response in practice. (2) In the actual vibration process, the damping ratio of the bridge usually changes with the amplitude, showing a nonlinear characteristic. However, in our 2D numerical simulation, the damping ratio is treated as a fixed value, which is different from the actual situation. (3) The 2D section method ignores the three-dimensional effect of the segment model along the span direction in the experiment, which may lead to the relative amplification of the predicted amplitude.

4.4. Flow field characteristic analysis

The cross-sectional geometry of double-layer steel truss girders is complex, with diverse bridge deck components. Variations in turbulence characteristics among these components may induce the formation of vortices with differing sizes and shedding frequencies [33,34]. To identify the vortex regions driving the VIV of the main girder section, the completed bridge section was studied under $+3^\circ$. Twenty-six monitoring points were strategically placed at various locations in the flow field to record wind speed variations, as shown in Fig. 17.

Above the upper deck were monitoring points P1-P6, below it were P8-P13, and behind it was P7. Behind the two diagonal braces were points P14 and P15, respectively. Points P16-P18 were located above the lower deck, P21-P25 below the lower deck, and P19 behind the lower deck. Points P20 and P26 were positioned behind the maintenance vehicle tracks, respectively. Wind speed time series were recorded at each monitoring point, and Fourier transform analysis was applied to the frequency-domain signals to obtain the characteristic frequencies and corresponding Strouhal numbers for the 26 monitoring points, as presented in Table 3.

Table 3 indicates that, under $+3^\circ$, when the main girder section undergoes vertical bending VIV, the wind tunnel experiment on the segment model yields a St number of 0.47. Similar St values of 0.49 were recorded at monitoring points P1-P7, implying VIV stems from vortex shedding occurring in two key areas: above the upper deck and in the wake. When the main girder section undergoes torsional VIV, the wind tunnel experiment on the segment model yields St numbers of 0.36 and 0.64. Similar St values of 0.33 and 0.65 were recorded at monitoring point P19, indicating that the VIV is primarily driven by vortex shedding in the lower deck's wake zone. When the main girder's aerodynamic shape alters, careful analysis of airflow changes above the upper deck, in its wake, and in the lower deck's wake is essential.

5. Analysis of the influence of railing ventilation rate on the VIV mechanisms of double-layer steel truss girder sections

This section aims to systematically analyze the influence of different types of railings under various ventilation rate conditions on the VIV response of the double-deck steel truss girder section, and to reveal the aerodynamic mechanisms that dominate the vertical bending and torsional VIV. The analysis will be unfolded in two parts: first, focusing on the upper railings (maintenance walkway railing and upper crash railing), by analyzing the instantaneous vorticity evolution and spectral characteristics of the surrounding flow field, to reveal how they induce flow separation, enhance vortex shedding strength, and ultimately

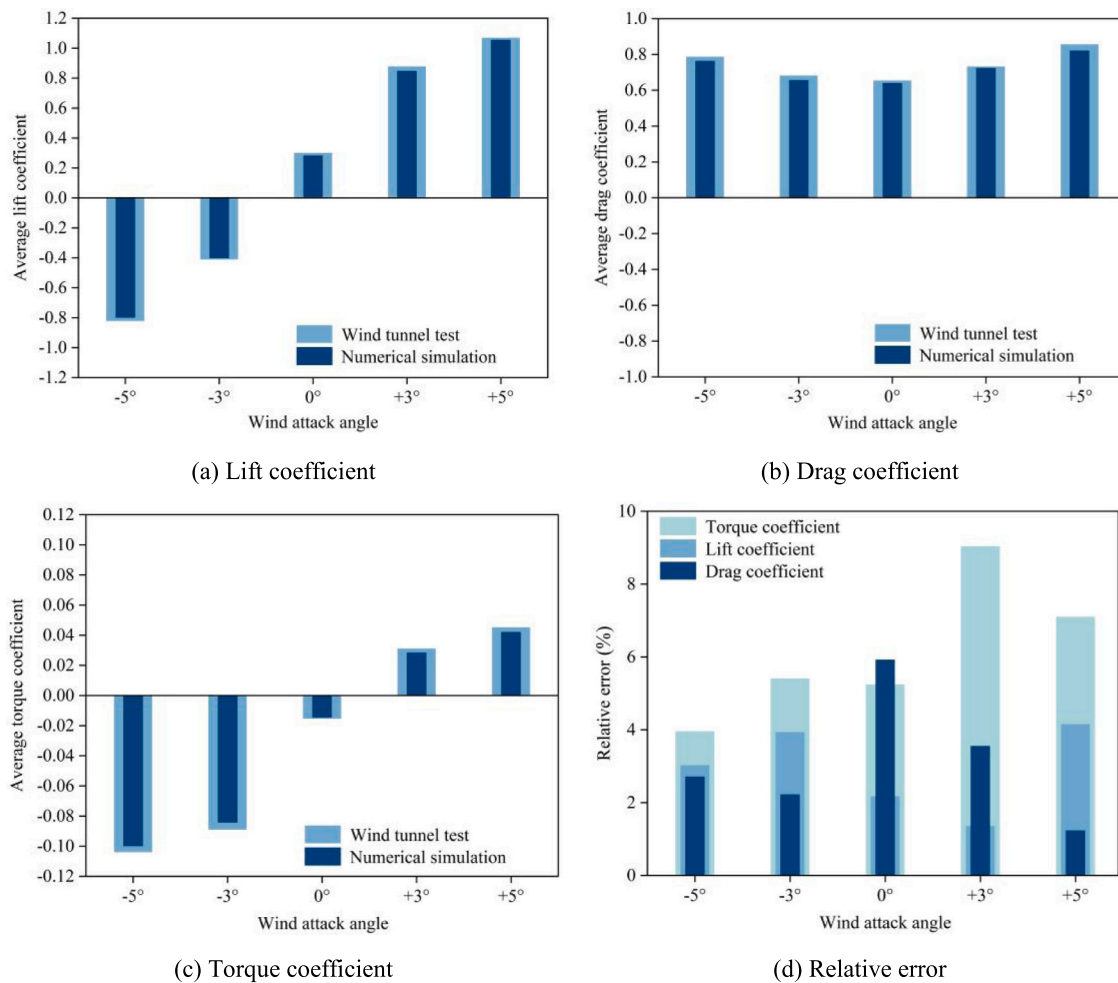


Fig. 15. Comparison of force measurement test and simulation results.

excite the aerodynamic mechanism of the main girder's vertical bending VIV response. Subsequently, a further analysis will be conducted on the lower railings (pedestrian railing and lower crash railing), to deeply investigate how they, in turn, suppress the intrinsic mechanism of the main girder's torsional VIV response by changing the vortex evolution patterns in the wake region and weakening the unsteady aerodynamic forces.

5.1. Analysis of the influence mechanisms of maintenance walkway railings with varying ventilation rates

To further investigate the excitation mechanism of reducing the ventilation rate of maintenance walkway railings on the vertical bending VIV of the main girder section, flow field visualization and spectral analysis were performed for three conditions. For three ventilation rate conditions, the peak vertical bending VIV response at +3° was analyzed, with the temporal evolution of instantaneous vorticity in the upper deck region of the main girder over one cycle shown in Fig. 18.

Fig. 18 shows that, compared to the instantaneous vorticity changes in the upper deck region of the original section, reducing maintenance walkway railings ventilation rate alters the flow field distribution characteristics in this region. Lower ventilation rates enhance the wind-blocking effect of maintenance walkway railings, making it harder for oncoming airflow to pass through the railing gaps, restricted flow separation forces more airflow upward along the tops of maintenance walkway railings, increasing local airflow velocity in this region, the flow shear layer intensifies, inducing larger-scale vortex shedding and forming a broader wake region. The stronger obstruction and diversion

of airflow caused by reduced ventilation of maintenance walkway railings directly increases the aerodynamic forces on the main girder section, particularly the periodic aerodynamic excitation forces closely related to vortex shedding. Consequently, the vortex-induced forces acting on the upper deck region are enhanced.

Fig. 19 presents the frequency spectra for wind speed time-series data at points P3 above the upper deck and P7 in the wake, corresponding to the peak vertical bending VIV response of the main girder.

As shown in Fig. 19(a), analysis indicates that during vertical bending VIV, the vortex shedding frequency at the tops of maintenance walkway railings closely matches the vertical bending fundamental frequency of the test model, suggesting a resonant coupling effect between vortex shedding and structural vibration. As maintenance walkway railings ventilation rate decreases, the amplitude of low-frequency components associated with larger-scale vortex shedding from the railing tops gradually increases, indicating enhanced vortex strength. The amplitude of high-frequency components associated with small-scale vortex shedding from maintenance walkway railings gaps gradually decreases to zero, indicating suppression and dissipation of small-scale vortices. This change increases the dominant scale of shed vortices and makes the vortex shedding process more regular and stable. Larger-scale and more regular vortex shedding generates stronger periodic aerodynamic excitation forces on the main girder, enhancing vortex-induced forces and significantly increasing the amplitude of the vertical bending VIV response. Furthermore, enhanced stability in vortex shedding frequency promotes resonance by facilitating lock-in with the structural natural frequency. This phenomenon consequently reduces the critical wind speed for onset, widens the lock-in velocity band,

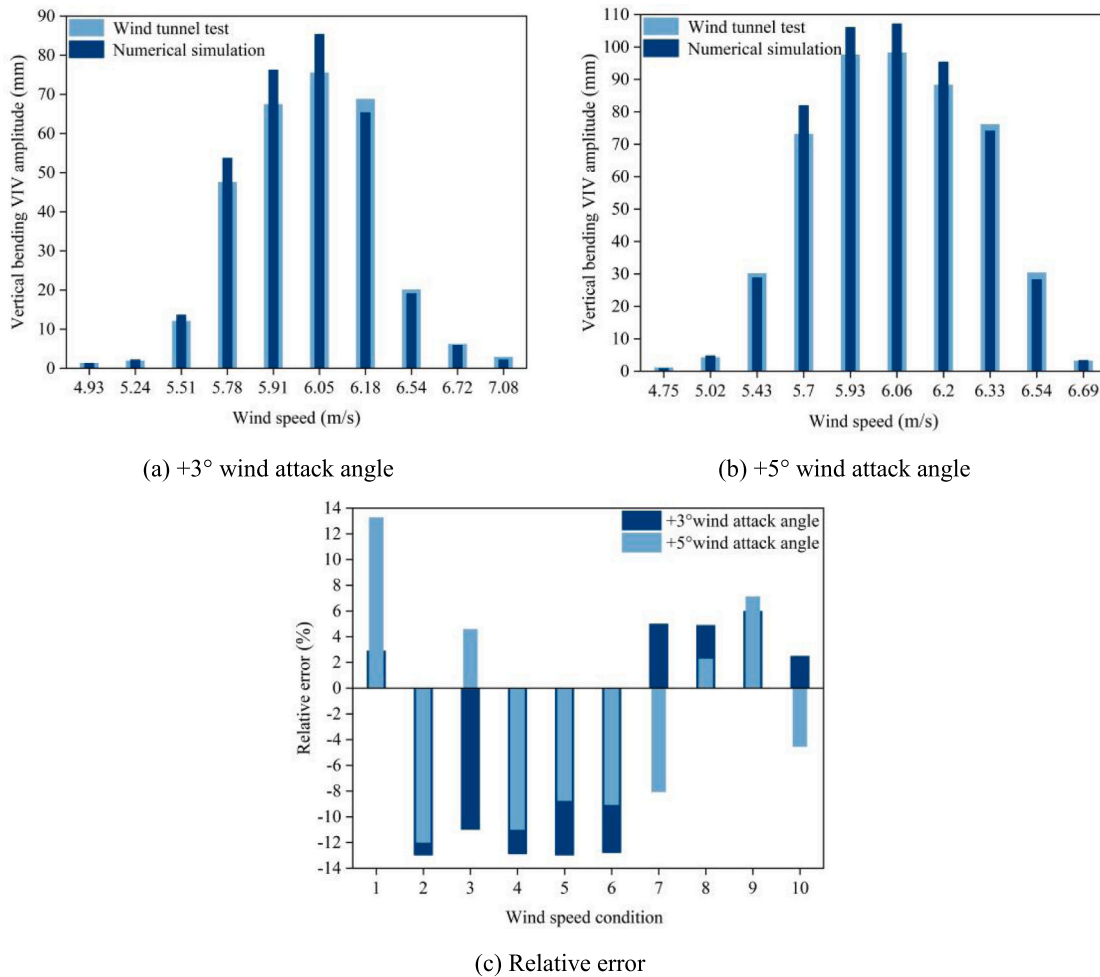


Fig. 16. Comparison of vibration measurement test and simulation results.

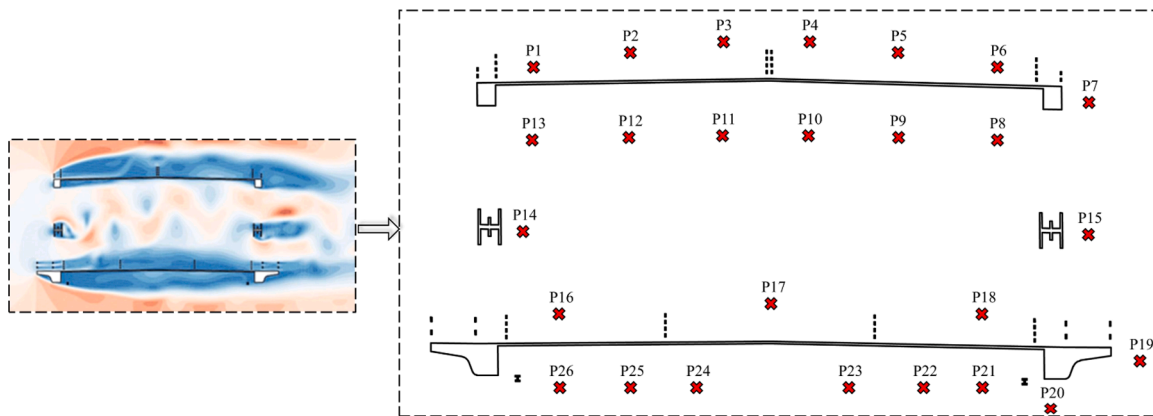


Fig. 17. Flow field monitoring point arrangement.

and elevates the susceptibility of bridges to vertical bending VIV across a broader spectrum of lower wind speeds.

Fig. 19(b) indicates the consistent amplitude of the characteristic frequency component despite decreasing maintenance walkway railings ventilation rate. This is likely because, under +3° wind attack angle, the windward deck and railings obstruct airflow, preventing flow over the leeward side above the upper deck. Consequently, altering the railing ventilation rate has minimal impact on the airflow characteristics in this region. This stability ensures the vortex shedding pattern in the upper

deck's wake remains unaffected, thus having no influence on the main girder's vertical bending VIV. Overall, spectral analysis corroborates that decreased ventilation in the maintenance walkway railing impairs the main girder section's aerodynamic performance. This enhances vortex strength at the top of the windward railings and increases vortex scale, thereby subjecting the upper deck to stronger vortex-induced forces. This aerodynamic mechanism primarily drives the significant increase in the main girder section's vertical bending VIV response.

Table 3
Monitoring point peak characteristic analysis.

Monitoring point	Vertical bending VIV frequency	Torsional VIV frequency	Strouhal number
P19	1.55 Hz, 3.06 Hz	3.93 Hz, 7.74 Hz	0.33, 0.65
P1-P7	2.31 Hz	5.83 Hz	0.49
P8-P13, P20-P26	2.73 Hz	6.91 Hz	0.58
P1-P6, P16-P18	4.29 Hz	10.83 Hz	0.91
P20, P26	5.09 Hz	12.86 Hz	1.08
P15	8.67 Hz	21.91 Hz	1.84
P7-P14	10.23 Hz	25.84 Hz	2.17

5.2. Analysis of the influence mechanisms of upper crash railings with varying ventilation rates

This section further investigates the excitation mechanism by which reducing the ventilation rate of the upper crash railing influences the main girder's vertical bending VIV response. Focusing on the peak vertical bending VIV response point under $+3^\circ$, the instantaneous vorticity evolution in the upper deck region of the main girder over a single vibration cycle, following changes in the upper crash railing ventilation rate, is depicted in Fig. 20.

Fig. 20 indicates that, at $+3^\circ$ wind attack angle, decreased ventilation in the upper crash railing leads to marked changes in the airflow characteristics above the upper deck. However, below the deck, the airflow undergoes significant reattachment (flow regeneration). Obstructed by the windward upper chord, the flow quickly bends down, adhering to the upper deck's underside. The flow characteristics in this region show no significant variation with changes in the railing ventilation rate.

Fig. 21 displays the spectral phase measured at two monitoring points: P3, situated above the upper deck, and P7, within the wake region. These measurements were taken when the main girder reached its peak vertical bending VIV response.

Spectral analysis results in Fig. 21(a) indicate that, similar to conditions with the maintenance walkway railing ventilation rate, when the upper crash railing ventilation rate is high, a portion of the incoming flow can pass smoothly through the railing gaps. Consequently, the airflow kinetic energy in the region above the railing is relatively low, resulting in a weaker active disturbance to the flow field. As the railing ventilation rate decreases, the combined obstruction effect of the windward upper chord and railings intensifies, making it difficult for airflow to pass directly through the railing gaps. This enhances the recirculation strength between the maintenance walkway railing and the upper crash railing, forcing the airflow to move upward along the railings. Consequently, the airflow intensity at the top edge of the railings increases, forming a broader wake and inducing stronger vortex shedding. The increased vortex strength directly impairs the aerodynamic stability of the main girder section, resulting in a substantial increase in the vertical bending VIV response. This amplifies the amplitude of low-frequency components associated with large-scale vortices, while the amplitude of high-frequency components linked to small-scale gap flows diminishes and approaches zero. Furthermore, reducing the ventilation rate of the upper crash railing results in an observed increase in vortex shedding frequency. This implies that, at lower wind speeds, the shedding frequency can reach the vertical bending VIV lock-in frequency, leading to an earlier onset wind speed, an expanded lock-in range, and a heightened risk of vertical bending VIV.

Fig. 21(b) highlights a contrast: while flow conditions at the windward railing top may vary, the vortex shedding pattern in the leeward upper deck wake remains largely unaffected by reduced railing ventilation. The flow characteristics in this region remain stable, suggesting that its contribution to the main girder's vertical bending VIV of is

relatively independent of changes in the upper crash railing ventilation rate and has a minor impact. The enhanced vortex strength at the top of the windward upper crash railing is the primary factor driving the significant intensification of the main girder's vertical bending VIV response. Overall, decreased ventilation in the upper crash railing modifies the main girder section's aerodynamic profile, shifting it from a relatively permeable open design toward a more closed, bluff shape. This enhances the overall vortex shedding effect of the section, particularly increasing the strength and scale of vortices generated at the top of the windward upper crash railing, thereby heightening the main girder's VIV susceptibility and inducing a stronger vertical bending VIV response.

5.3. Analysis of the influence mechanisms of lower pedestrian railings with varying ventilation rates

To delve deeper into how decreased ventilation in the pedestrian railing suppresses the main girder section's torsional VIV response, flow field visualization and spectral analysis were conducted for the main girder section under four ventilation rate conditions. Focusing on the peak torsional VIV response point under $+3^\circ$, the instantaneous vorticity evolution in the lower deck's wake over one vibration cycle for the four ventilation rate conditions is depicted in Fig. 22. Due to the significant difference in the critical wind speeds for torsional and vertical bending VIV, their corresponding vorticity magnitudes also vary considerably. Therefore, to clearly visualize the instantaneous vorticity evolution process under each condition, different legend scales are used for the vorticity contour plots in this and subsequent sections.

Higher ventilation rates enhance flow mobility, resulting in larger and stronger vortex separation and shedding. Fig. 22 indicates that, relative to the original section's instantaneous wake vorticity evolution, reduced pedestrian railing ventilation at $+3^\circ$ significantly disrupts airflow characteristics within the lower deck's wake. As the ventilation rate decreases, the airflow through the railing gaps diminishes, causing the vortices generated by gap flows to weaken and reduce in number until, with fully impermeable railings, small-scale gap vortices disappear entirely. Meanwhile, the size of vortices induced at the top of the pedestrian railing and the surface of the lower chord remains largely unchanged, but their strength exhibits a clear downward trend. Reduced pedestrian railing ventilation leads to a lower vortex shedding frequency. This necessitates higher wind speeds to achieve locking with the torsional VIV frequency, consequently delaying the onset wind speed. The flow-rectifying effect of the lower crash railing at the trailing edge of the lower deck reduces the aerodynamic drag on the main girder, weakening the vortex-induced forces acting on the lower deck.

Fig. 23 presents the frequency spectrum corresponding to the wind speed time series recorded at P19 in the lower deck's wake at the peak torsional VIV response point of the main girder.

Spectral analysis from Fig. 23 further quantifies the changes in vortex shedding frequency and strength. The wind speed time series from P19 is primarily composed of two dominant frequency components: the higher frequency component corresponds to small-scale vortices generated at the gaps of the pedestrian railing, while the lower frequency component corresponds to large-scale vortices produced at the top of the pedestrian railing and the surface of the lower chord. At the secondary torsional VIV peak response point of the main girder, the vortex shedding frequency in the maintenance walkway railing gap region, and at the primary torsional VIV peak response point, the vortex shedding frequencies at the railing tops and lower chord surface region, closely match the torsional fundamental frequency of the test model, indicating their role as primary drivers of main girder torsional VIV. Reduced pedestrian railing ventilation leads to a gradual decrease in the amplitudes of the two dominant frequency components, indicating weakened dominant vortices. When airtight railings are used, high-frequency components vanish, while low-frequency components persist. This is likely because vortex shedding from airflow through

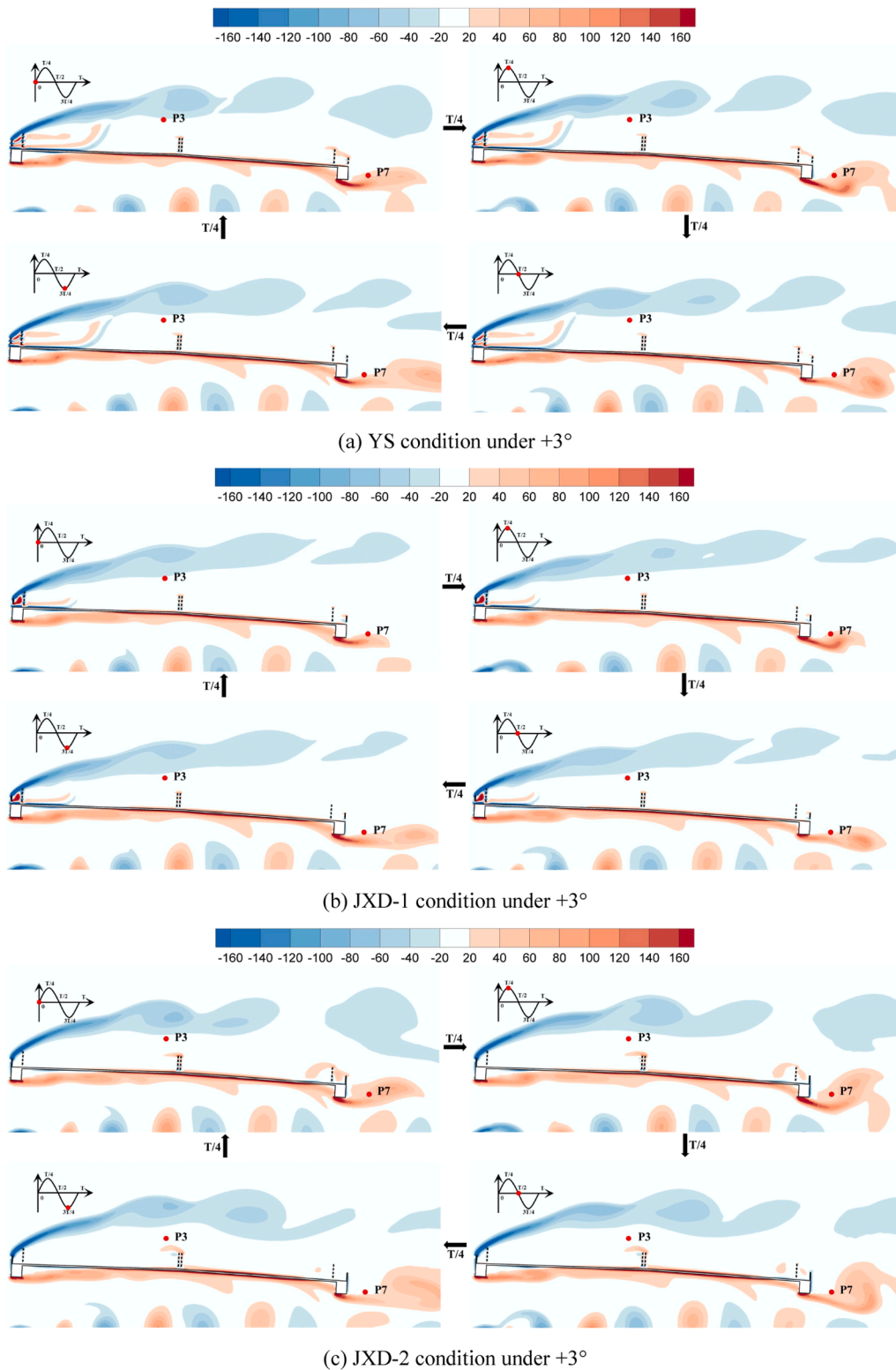


Fig. 18. Vorticity evolution on the upper deck and wake region under, maintenance walkway railing ventilation rates.

upper crash railing gaps also affects main girder torsional VIV, but was not recorded due to the P19 monitoring point's location. Overall, reducing pedestrian railing ventilation significantly mitigates the structure's VIV response by altering the flow field below the main girder section. This alteration weakens vortex shedding near the leeward pathway, dissipates small vortices, and decreases vortex-induced forces on the lower deck.

5.4. Analysis of the influence mechanisms of lower crash railings with varying ventilation rates

To further investigate the mechanism by which reducing the ventilation rate of lower crash railings suppresses torsional VIV in the main girder section, flow field visualization and spectral analysis were conducted for three conditions. At the peak torsional VIV response point

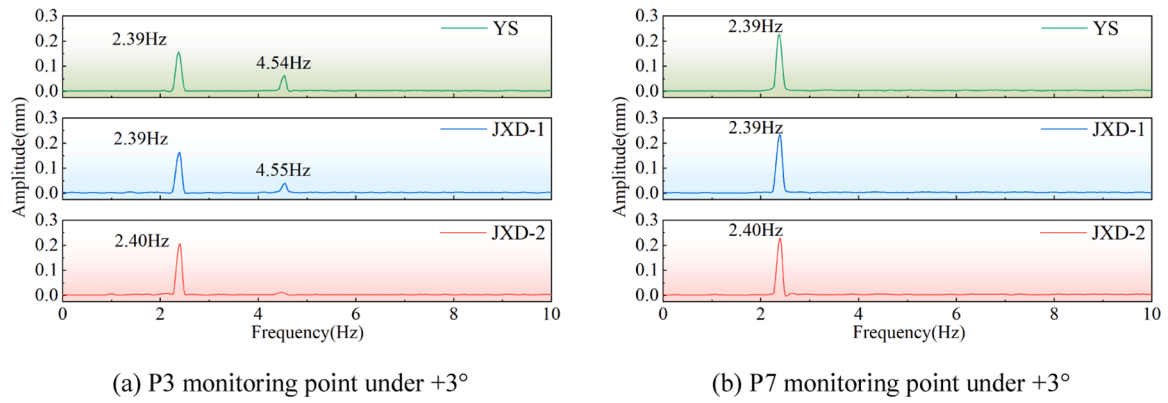


Fig. 19. Spectral phase at P3 and P7 under maintenance walkway railing ventilation rates.

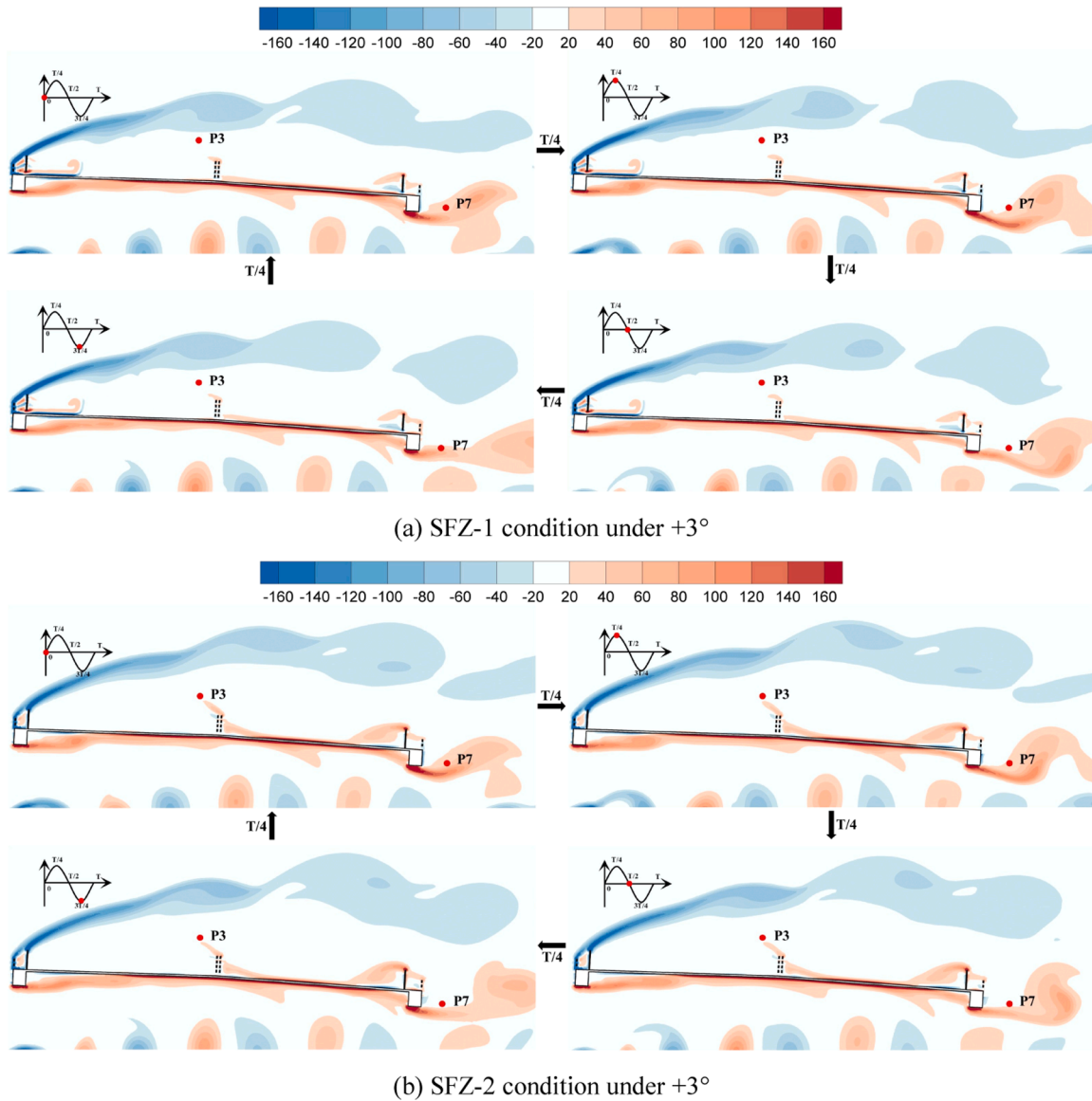


Fig. 20. Vorticity evolution on the upper deck and wake region under, upper crash railing ventilation rates.

under $+3^\circ$, the temporal evolution of instantaneous vorticity in the lower deck's wake over one cycle, for three railing ventilation rates, is shown in Fig. 24.

Fig. 24 illustrates energy dissipation in the oncoming flow above the

lower deck due to lower crash railings, mirroring the effect seen with pedestrian railing ventilation. This dissipation leads to vortex separation, shedding, and fragmentation. At the main girder's peak torsional VIV response, Fig. 25 displays the spectral phase measured at

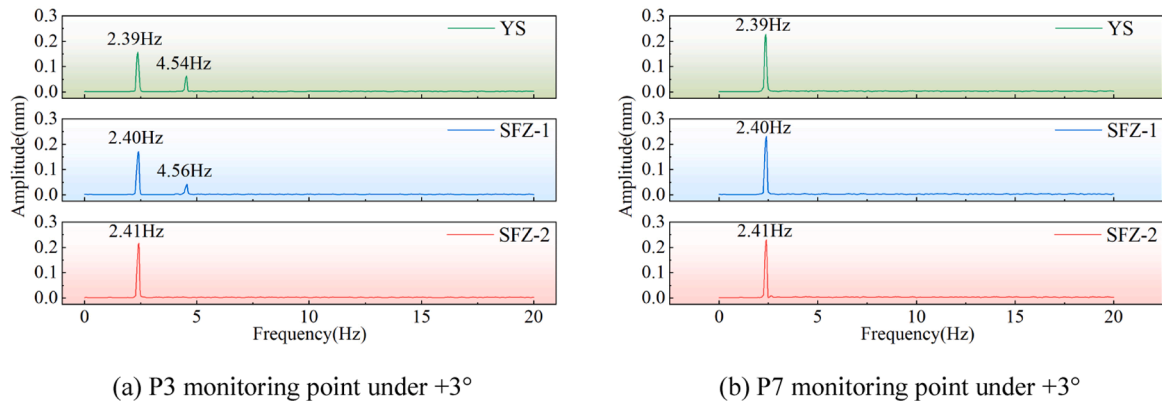


Fig. 21. Spectral phase at P3 and P7 under upper crash railing ventilation rates.

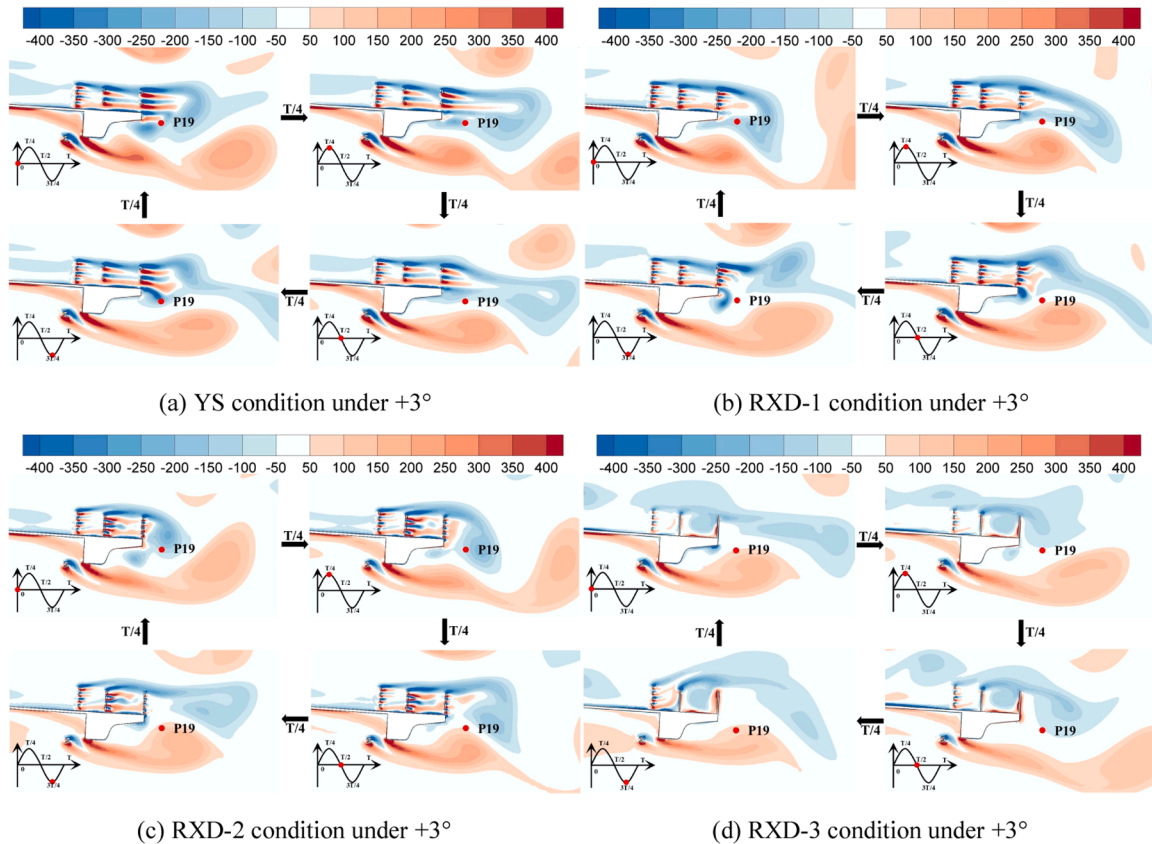


Fig. 22. Vorticity evolution on the upper deck and wake region under, pedestrian railing ventilation rates.

monitoring point P19 within the lower deck's wake.

Fig. 25 shows that, compared to the YS condition, reducing the ventilation rate of lower crash railings significantly decreases the amplitudes of dominant frequency components, indicating a substantial reduction in the strength of vortices induced by the lower railings. As the ventilation rate decreases, airflow struggles to pass through the railing gaps, altering the original vortex shedding pattern. Although the size of vortices shed from the tops of lower crash railings and the lower chord surface remains largely unchanged, their strength significantly decreases, vortex asymmetry diminishes, and the amplitude of bridge aerodynamic moment fluctuations reduces, effectively suppressing the main girder's torsional VIV response. In conclusion, reduced ventilation in lower crash railings effectively mitigates the main girder section's VIV response, especially torsional VIV, by substantially optimizing its aerodynamic performance.

6. Conclusions

Leveraging wind tunnel tests and numerical simulations, this study systematically examined how varying the ventilation rate of the four railing types (maintenance walkway railings, upper crash railings, lower pedestrian railings, and lower crash railings) influences the double-layer steel truss girder section's VIV performance and associated flow field characteristics at $+3^\circ$. The main conclusions are as follows:

- (1) The wind tunnel test results indicate that the wind attack angle has a significant effect on the VIV performance of the section model. Significant VIV phenomena were observed on the test model only at $+3^\circ$, while none were detected at -3° and 0° . Under the $+3^\circ$ condition where VIV occurred, the lock-in range for the vertical bending vibration corresponded to a full-scale

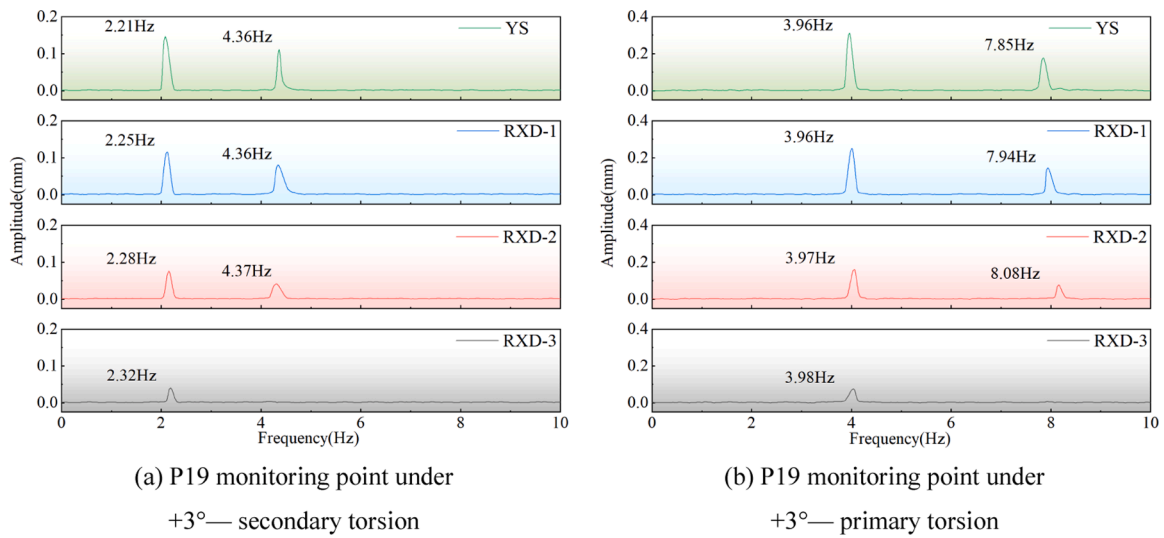


Fig. 23. Spectral phase at P19 under pedestrian railing ventilation rates.

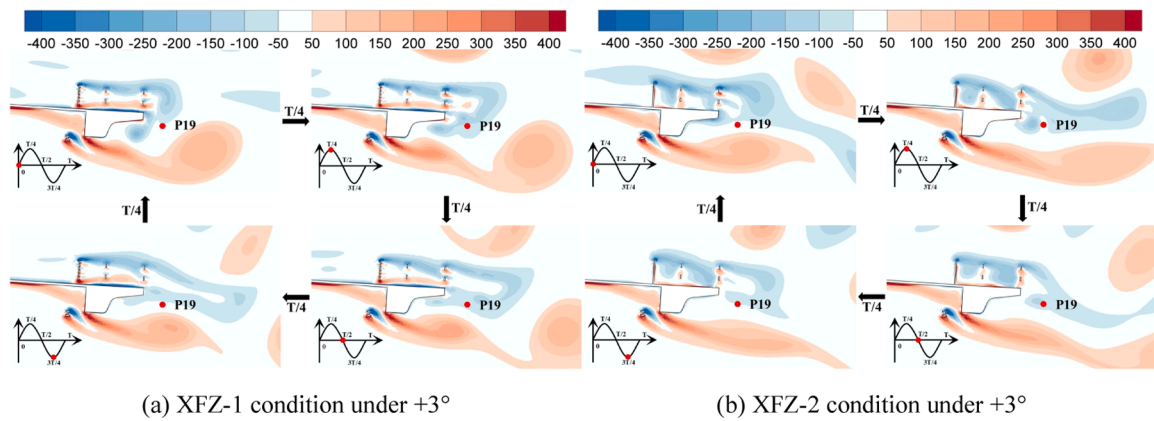


Fig. 24. Vorticity evolution on the upper deck and wake region under, lower crash railing ventilation rates.

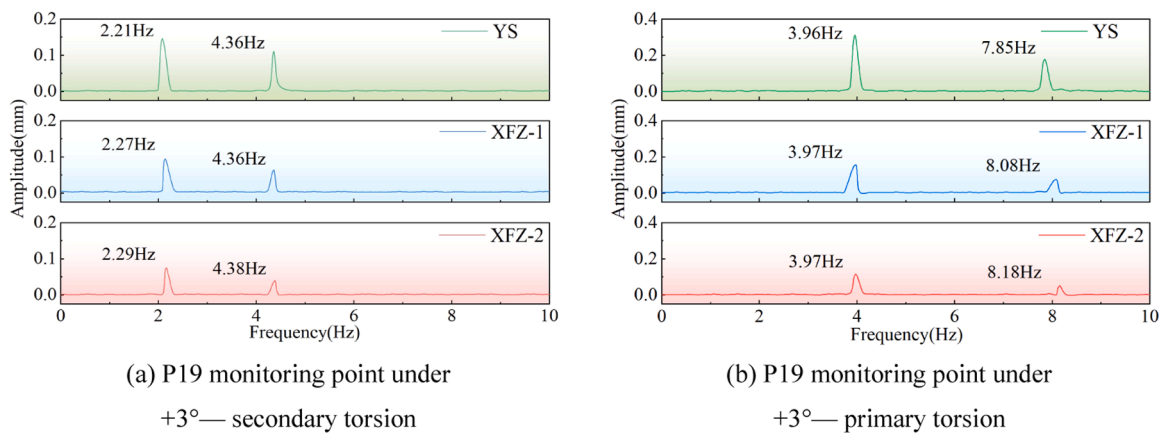


Fig. 25. Spectral phase at P19 under lower crash railing ventilation rates.

wind speed of 4.93–6.72 m/s, with a maximum vertical bending amplitude of 75.54 mm (corresponding to a full-scale wind speed of 6.05 m/s). The torsional VIV at this attack angle included two lock-in regions: a secondary torsional VIV with a wind speed lock-in range of 8.13–9.52 m/s, and a primary torsional VIV with a wind speed lock-in range of 12.73–18.06 m/s. Their maximum

- amplitudes were 0.079° (at a full-scale wind speed of 8.60 m/s) and 0.181° (at a full-scale wind speed of 15.28 m/s), respectively.
- (2) Reducing the ventilation rate of maintenance walkway railings or upper crash railings amplifies both the vertical bending and torsional VIV responses of the main girder, with the excitation effect on vertical bending VIV being particularly pronounced at 0 % ventilation rate for upper crash railings, resulting in a 33 %

amplitude increase. Reducing the ventilation rate of lower pedestrian railings or lower crash railings effectively suppresses the main girder's torsional VIV response, with the optimal suppression achieved at 0 % ventilation rate for pedestrian railings, yielding a 78.5 % suppression rate for secondary torsional VIV and a 71.3 % suppression rate for primary torsional VIV.

- (3) The primary excitation mechanism of reducing the ventilation rate of maintenance walkway railings or upper crash railings lies in restricting airflow through railing gaps, intensifying the local stagnation effect of the oncoming flow at the leading edge of the upper deck. This induces strong flow separation at the railing tops, forming high-intensity vortices. These vortices, shedding at high frequency, evolve toward the leeward side, enhancing flow field turbulence in the upper deck region, significantly increasing the vertical bending vortex-induced forces on the main girder, and markedly amplifying the vertical bending VIV response.
- (4) Reducing lower pedestrian railings ventilation or lower crash railings suppresses torsional VIV primarily by effectively inhibiting the formation and shedding of small-scale vortices at railing gaps, while significantly reducing the strength and asymmetry of vortices shed from railing tops and the lower chord. This markedly decreases wake vorticity in the lower deck region, diminishing aerodynamic disturbances near the lower deck. As torsional VIV is primarily driven by uneven airflow distribution across the lower deck, reduced wake vorticity mitigates aerodynamic asymmetry on both sides of the main girder, thereby lowering the torsional VIV driving force and significantly attenuating the torsional VIV response.

CRediT authorship contribution statement

Gang Yao: Writing – review & editing, Methodology, Conceptualization. **Yang Yang:** Writing – review & editing, Methodology, Conceptualization. **Huanhuan Li:** Writing – original draft, Software, Formal analysis. **Long Li:** Writing – original draft. **Junhao Yang:** Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Guo J, Tang H, Li Y, Wang Z. Effects of guardrails on wind environment for vehicles and aerodynamic stability for bridges with box girders. *Adv Struct Eng* 2021;24: 453–69. <https://doi.org/10.1177/1369433220956827>.
- [2] Wu F, Cui W, Zhao L, Guo Z, Li Q, Ge Y. Field measurements of wind microclimate at the vehicle level on a bridge deck under typical canyon terrain. *J Bridge Eng* 2023;28:4023013. <https://doi.org/10.1061/JBENF2.BEENG-5875>.
- [3] Zhang M, Jiang X, Zhang J, Zhang J, Qin J, Li Y. Field-measurement-based local wind field characteristics on bridge deck of truss girder bridges in a deep-cutting gorge bridge site area. *Meas Control* 2023;56:668–80. <https://doi.org/10.1177/00202940221105866>.
- [4] Nagao F, Utsunomiya H, Yoshioka E, Ikeuchi A, Kobayashi H. Effects of handrails on separated shear flow and vortex-induced oscillation. *J Wind Eng Ind Aerodyn* 1997;69–71:819–27. [https://doi.org/10.1016/S0167-6105\(97\)00208-0](https://doi.org/10.1016/S0167-6105(97)00208-0).
- [5] Zhong Y, Li M, Ma C. Aerodynamic admittance of truss girder in homogenous turbulence. *J Wind Eng Ind Aerodyn* 2018;172:152–63. <https://doi.org/10.1016/j.jweia.2017.10.019>.
- [6] Ge Y, Zhao L, Cao J. Case study of vortex-induced vibration and mitigation mechanism for a long-span suspension bridge. *J Wind Eng Ind Aerodyn* 2022;220: 104866. <https://doi.org/10.1016/j.jweia.2021.104866>.
- [7] Bai H, Ji N, Xu G, Li J. An alternative aerodynamic mitigation measure for improving bridge flutter and vortex induced vibration (VIV) stability: sealed traffic barrier. *J Wind Eng Ind Aerodyn* 2020;206:104302. <https://doi.org/10.1016/j.jweia.2020.104302>.
- [8] Cui X, Gao L, Liu J. Wind tunnel test study on the influence of railing ventilation rate on the vortex vibration characteristics of the main beam. *Int J Heat Technol* 2018;36:65–71. <https://doi.org/10.18280/ijht.360109>.
- [9] Liu J, Xian B, Li F, Tian L, Qiu E, Ni Z. Optimizing vortex-induced vibration suppression in π -shaped girders using wind tunnel testing and CFD analysis. *Adv Struct Eng* 2025;28:1010–28. <https://doi.org/10.1177/13694332241300671>.
- [10] Wu Y, Wu X, Li J, Xin H, Sun Q, Wang J. Investigation of vortex-induced vibration of a cable-stayed bridge without backstays based on wind tunnel tests. *Eng Struct* 2022;250:113436. <https://doi.org/10.1016/j.engstruct.2021.113436>.
- [11] Liu L, Zou Y, He X, Yang J, Wang Z, Guo D. Experimental investigation on vortex-induced vibration of a long-span rail-cum-road bridge with twin separated parallel decks. *J Wind Eng Ind Aerodyn* 2022;228:105086. <https://doi.org/10.1016/j.jweia.2022.105086>.
- [12] Yao G, Chen Y, Yang Y, Zheng Y, Wu L, Du H. Research on mechanism of vortex-induced vibration railing effect of double-deck large-span suspension bridge. *Appl Sci* 2023;13:9314. <https://doi.org/10.3390/app13169314>.
- [13] Yan Y, Yagi T. Effects of handrail details on vortex-induced vibration for a box-girder bridge. *J Bridge Eng* 2022;27:4021114. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001835](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001835).
- [14] Li M, Sun Y, Jing H, Li M. Vortex-induced vibration optimization of a wide streamline box girder by wind tunnel test. *KSCE J Civ Eng* 2018;22:5143–53. <https://doi.org/10.1007/s12205-018-0548-y>.
- [15] Yang Y, Li L, Yao G, Wang M, Zhou C, Lei T, et al. Study on the influence of the fully enclosed barrier on the vortex-induced vibration performance of a long-span highway–railway double-deck truss bridge. *Phys Fluids* 2024;36:087132. <https://doi.org/10.1063/5.0220758>.
- [16] Shang C, Xiang H, Bao Y, Li Y, Luo K. Mechanism of vortex-induced vibration in two lock-in regions for truss girder sections. *J Wind Eng Ind Aerodyn* 2024;255: 105946. <https://doi.org/10.1016/j.jweia.2024.105946>.
- [17] Yang H, Zheng S, Ma C. Vertical vortex-induced vibration performance of split triple-box girders with various ancillary facility configurations. *Phys Fluids* 2025; 37:27128. <https://doi.org/10.1063/5.0254403>.
- [18] He Z, Zhao Y, Zhang H, Tang H, Zhu Q, Ai Y, et al. Vortex-induced vibrations and galloping of a square cylinder: the impact of damping and mass ratio. *Ocean Eng* 2025;320:120371. <https://doi.org/10.1016/j.oceaneng.2025.120371>.
- [19] Tang H, Kang J, Li Y. Aerostatic and aerodynamic stability of a suspension bridge during early erection stages. *J Wind Eng Ind Aerodyn* 2023;233:105297. <https://doi.org/10.1016/j.jweia.2022.105297>.
- [20] Fang C, Hu R, Tang H, Li Y, Wang Z. Experimental and numerical study on vortex-induced vibration of a truss girder with two decks. *Adv Struct Eng* 2021;24: 841–55. <https://doi.org/10.1177/1369433220969026>.
- [21] Xin D, Zhan J, Zhang H, Ou J. Control of vortex-induced vibration of a long-span bridge by inclined railings. *J Bridge Eng* 2021;26:4021093. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001803](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001803).
- [22] Yao G, Wu L, Yang Y, Liu M, Chen Y, Du H. Study on the suppression effect on vortex-induced vibration of double-deck truss girder by the spatial position of the deflector plate. *Appl Sci* 2023;13:9764. <https://doi.org/10.3390/app13179764>.
- [23] Liu X, Xu F, Hou Y, Duan Z, Ou J. Numerical study on passive-jet flow control for chamfered square cylinder. *Ocean Eng* 2024;295:116860. <https://doi.org/10.1016/j.oceaneng.2024.116860>.
- [24] Ishihara T, Li T. Numerical study on suppression of vortex-induced vibration of circular cylinder by helical wires. *J Wind Eng Ind Aerodyn* 2020;197:104081. <https://doi.org/10.1016/j.jweia.2019.104081>.
- [25] Li T, Ishihara T. Numerical study on vortex-induced vibration of circular cylinder with two-degree-of-freedom and geometrical nonlinear system. *J Fluids Struct* 2021;107:103415. <https://doi.org/10.1016/j.jfluidstruct.2021.103415>.
- [26] Yang Y, Li L, Yao G, Wu B, Wang D, Yu H, et al. Study on the influence of wind fairing parameters on the aerodynamic performance of long-span double-deck steel truss suspension bridge. *Buildings* 2024;14:2255. <https://doi.org/10.3390/buildings14072255>.
- [27] Yue H, Zhang H, Zhu Q, Ai Y, Tang H, Zhou L. Wake dynamics of a wind turbine under real-time varying inflow turbulence: a coherence mode perspective. *Energy Convers Manag* 2025;332:119729. <https://doi.org/10.1016/j.enconman.2025.119729>.
- [28] Qiao L, Zhao Y, Zhou L, Ai Y, Zhu Q, Zhang H. Coherence mode and floquet analysis on flow past a rectangular cylinder with small angle of attack. *Phys Fluids* 2024;36:123616. <https://doi.org/10.1063/5.0241201>.
- [29] Hu W, Liu J, Liu Y, Lyu Y, Li H, Xian J, et al. Construction alignment and closure control of CFST truss arch bridges based on temperature effect. *Structures* 2024;63: 106471. <https://doi.org/10.1016/j.istruc.2024.106471>.
- [30] Ma Y, Yang Z, Liu Y, Liu J, Zhao W, Qiu R, et al. Flexural behavior of rectangular concrete filled steel tubular (RCFST) composite truss bridge. *Structures* 2025;72: 108260. <https://doi.org/10.1016/j.istruc.2025.108260>.
- [31] Xu F, Li D, Gao P, Zang W, Duan Z, Ou J. Numerical simulation of two-dimensional transmission line icing and analysis of factors that influence icing. *J Fluids Struct* 2023;118:103858. <https://doi.org/10.1016/j.jfluidstruct.2023.103858>.

- [32] Yao G, Chen Y, Yang Y, Zheng Y, Du H, Wu L. Investigation of vortex-induced vibration of double-deck truss girder with aerodynamic mitigation measures. *J Mar Sci Eng* 2023;11:1118. <https://doi.org/10.3390/jmse11061118>.
- [33] Wu B, Zhou J, Li S, Xin J, Zhang H, Yang X. Combining active and passive wind tunnel tests to determine the aerodynamic admittances of a bridge girder. *J Wind Eng Ind Aerodyn* 2022;231:105180. <https://doi.org/10.1016/j.jweia.2022.105180>.
- [34] Gang Y, Yang Y, Bo W, Lianjie L, Liangliang Z. Aerodynamic admittance influence on buffeting performance of suspension bridge with streamlined deck. *J Vibroeng* 2019;21:198–214. <https://doi.org/10.21595/jve.2018.19681>.